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The ESG Performance in Relation to the Board of Directors' Characteristics and Sustainability Incentives: A Cross-National Study

TIJANI AMARA¹ and ALI AHMADI²

¹ Assist Prof Accounting. Director, Higher Institute of Business Administration, Department of Accounting and Tax, University of Gafsa, Tunisia. e-mail: tijani_amara@yahoo.fr

² Assist Prof Accounting. Higher Institute of Business Administration, Department of Accounting and Tax, University of Gafsa, Tunisia. e-mail: ahmadi2402@gmail.com

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ABSTRACT

This paper seeks to find the impact of board of directors characteristics', compensation policy, and the CSR committee on the ESG performance by examining the influence of board characteristics', sustainability incentive compensation, and a CSR committee on ESG performance. By referring to one of the largest datasets to date, we use an unbalanced data set of 36,764 constant-year observations from 6000 listed companies, covering a 16-year period (2002-2017) from 60 countries around the world. Anyway, we show that the share of board independence is positively related to the ESG scores. There is an important effect of gender diversity in the board of directors on the firm performance' in the area of environmental, social and institutional governance. The results reveal that offsetting sustainability incentives has a positive effect in some geographic areas. There is a positive correlation between the ESG performance and the CSR Committee. Hypothesis 1. Board size positively affects ESG performance. Hypothesis 2. Board independence positively affects ESG performance. Hypothesis 3. A high level of gender diversity positively affects ESG performance. Hypothesis 4: ceteris paribus, large firms adopt sustainable ESG compensation incentives, which are greater for process-oriented ESG performance. Hypothesis 5. The establishment of a CSR sustainability committee has a positive impact on the performance of the ESG.

INTRODUCTION

Since it is responsible for approving and overseeing the implementation of strategic objectives, the governance system, and the creation of the company's culture, the company's performance depends critically on board of directors composition' (Merendino, & Melville, 2019). Also, a successful board of directors will emphasize business ethics and corporate responsibility. A growing number of studies has found a strong positive relationship between the company sustainability performance' and its profitability. Indeed, a company with higher sustainability ratings significantly outperform their peers, both in terms of market capitalization and accounting performance. In other words, adopting ESG best practices leads to a long-term competitive advantage.

This study contributes to both the academic literature and business practices in several aspects, firstly, it improves the current literature on the effect of board characteristics', sustainability incentive compensation and the CSR committee on ESG performance in a multinational context. Secondly, to our knowledge, current studies on banks analysing the relationship between ESG variables primarily address ESG disclosure (not ESG performance) typically focus on developing economies and ultimately cover an older time period. On the contrary, our empirical research covers the years 2002-2017 and analyses a large sample of 6000 listed companies in 60 countries, from different regions whose economies make a special contribution to sustainable development. Overall, all of our results contribute to the identification of banking best practices, particularly with regard to the composition of the company's board of directors, incentive bonuses in terms of sustainable development, and the CSR committee which are important drivers of the ESG outcome'.

Based on these considerations, this study aims to identify the characteristics of the board of directors that relate to the performance of environmental social governance. More specifically, according to the existing literature (Armstrong, 2020; Harjoto, & Wang, 2020) our analysis examines the effect of board characteristics (gender diversity, size, activity, compensation, and ESG controversies) on ESG performance in a sample of 36,764 observations from 60 countries from 2002 to 2017. Our focus on this sample allows us to have a more homogeneous reference group which is significant. Because these numbers of firms represent one of the major economic drivers of their economies which can promote the transition to a more inclusive and sustainable economy (Rajesh, 2020). To represent a company's sustainability performance, we use a Thomson Reuters Asset Score 4 ESG, which is a broad and documented measure of CSR performance that has been adopted by academics for both financial (Hübel, & Scholz, 2020; Kiriü, Nozaki, 2020) and non-financial (Hedqvist, & Larsson, 2020; Jahmane, & Gaies, 2020). Based on these considerations, this study aims to identify the characteristics of the board of directors that relate to the performance of environmental social governance.

1. LITERATURE BACKGROUND AND HYPOTHESES DEVELOPMENT

Our main empirical evidence shows that there is a linear relationship between the characteristics of the board of directors, the remuneration policy, the CSR committee and the ESG performance of the company. Interestingly, the correlation between the dependent and independent variables is linear confirming that the characteristics of the board of directors, the remuneration policy and the CSR committee positively influence the ESG performance of the company.

The variables most widely used in the literature to describe the impact of board size, gender diversity, board independence, sustainability incentives compensation and CSR committee on ESG performance. In the sub-sections below, we develop our hypotheses for each of these characteristics of corporate governance.

Board size

The size of the board of directors has mainly been studied in terms of its relation to the performance of the company. Empirical studies can be classified into two categories: a first current of research is in favour of small councils, while a second current of thought advocates large councils. Given that the companies in our sample are listed, and therefore are large and complex financial institutions operating in an intense regulatory environment, we expect larger boards to be more effective, especially in terms of workload and distribution of responsibilities, and greater diversity in terms of stakeholder representation, when and improving ESG performance (Zubeltzu-Jaka et al., 2020). Our hypothesis is consistent with findings from previous studies that show a positive relationship between board size and the extent of sustainability practices (Harjoto, & Wang, 2020). Based on these considerations, the following hypothesis is proposed for testing:

Hypothesis 1. Board size positively affects ESG performance.

Board independence

Within the context of using the ESG score for impression management, in this article we examine the existence of internal corporate governance control mechanisms that limit discretionary ESG scores that influence stakeholder perceptions about the sustainable performance of a company. Since the companies in our sample operate in a dynamic business environment, which requires an increased frequency of meetings to ensure legitimacy, legitimacy theory provides the basis for proposing a positive relationship between the frequency of board meetings and ESG performance. Our hypothesis is consistent with studies that find a positive relationship between board independence and ESG performance (Merendino, & Melville, 2019; Harjoto, & Wang, 2020). Concretely, and considering the main role of board independence on sustainable transparency, some authors have examined the role of board independence and ESG score. Several examples, analyzing a sample of companies, Merendino, & Melville, (2019) highlight high-impact management in stronger boards associated with greater independence of their members. Recently (Harjoto, & Wang, 2020) have documented that impression management depends on board characteristics, which influence the upbeat tone of ESG performance and subsequent performance. Based on the previous considerations, the following hypothesis is tested:

Hypothesis 2. Board independence positively affects ESG performance.

Women on the Board

Interpretations of the relationship between women on the board and ESG performance correlate with different traits of the women themselves. There appears to be a general consensus in the literature about the positive impact of female managers on sustainability performance. In more detail, women on boards positively influence company charitable contributions (Bettinelli et al., 2019; Abd Rahman et al., 2019), and climate change apprehension and reputation-based CSR metrics (Ajaz, et al., 2020). Similarly, several studies have found that gender diversity in the board increases the extent of social and environmental reporting and reduces environmental litigation. A possible explanation for these inconsistent results may be a nonlinear relationship between gender diversity in the board and social and environmental performance. Based on their findings, some studies (Ajaz et al., 2020) indicate that a higher level of women on the board improves ESG performance. Given this evidence, we develop the following hypothesis:

Hypothesis 3. A high level of gender diversity positively affects ESG performance

Executive compensation and carbon performance

Corporate governance has been identified as a determinant of managerial incentives for social performance, and several recent studies examine the role of executive compensation contracts that explicitly motivate managers for corporate social responsibility. Indeed, the moderating effect of a sustainable compensation policy based on ESG. Providing a sustainable compensation policy for ESG is likely to play a critical role in motivating corporate executives to take carbon initiatives, which in turn can enhance regulatory legitimacy. As part of sustainable corporate strategies, companies are increasingly using sustainable compensation policy to incentivize executives to pursue social and environmental projects. For example, Berndelli et al., (2018) found that sustainable compensation plans reduce the environmental exposure premium component of CEO compensation in different contexts. Therefore, we argue that with a sustainable compensation policy in place, the Board and the Board Compensation Committee may be in a better position to assess a company's carbon risk to design a more effective executive compensation plan, which may enhance ESG. Given this evidence, we develop the following hypothesis;

Hypothesis 4: *ceteris paribus*, large firms adopt sustainable ESG compensation incentives, which are greater for process-oriented ESG performance.

CSR Sustainability Committee

In recent years, companies have frequently opted to create a CSR committee to handle important sustainability duties. According to stakeholder theory, these committees typically assist the board of directors in overseeing corporate responsibility practices, but they can also play a key role in monitoring and evaluating CSR performance by ensuring compliance with ESG goals and strategies (Alessandrini, & Jondeau, 2020). In fact, the Board of Directors can design and implement CSR projects through the CSR committee to ensure enhanced performance of the ESG (Alessandrini, & Jondeau, 2020).

Furthermore, the CSR Committee is responsible for environmental and social information reporting procedures: it submits periodic reports to the Board of Directors on sustainability issues affecting the company, while also managing public disclosure of sustainability issues. For these reasons, the establishment of a CSR committee is seen as an important mechanism for an organization to maximize sustainable development opportunities. Recent studies such as Alessandrini, & Jondeau, (2020) suggests that the presence of CSR committees is important and positively correlated in achieving higher levels of ESG performance. For these motives, we postulate the following hypothesis:

Hypothesis 5. The establishment of a CSR sustainability committee has a positive impact on the performance of the ESG.

2. RESEARCH DESIGN

2.1 Sample Selection and Data Sources

Our initial sample was based on the availability of ESG data in the Thomson Reuters Asset4 and Thomson Reuters Eikon database. We started with the full available sample of 42,500 observations of non-financial companies listed on major stock exchanges in 60 countries listed in Panel A of Table 1. We then removed 5736 observations with missing I-level data on the independent variables used in our illustrative models. Our final sample is based on an unbalanced data set of 36,764 company year observations from 6000 listed companies, spanning a 16-year period (2002-2017). Table 1 shows the annual distribution by country for the sample. We used the Asset4 and Eikon dataset from Thomson Reuters to collect ESG data and corporate governance data. The Thomson Reuters ASSET4 database is used in our research because it is easily accessible to many investors and academics.

Table 1. Sample distribution

<i>Panel A. Distribution of the sample by country</i>		
<i>Firms</i>	<i>Number of observations per country</i>	<i>Percent (%) per country</i>
Argentina	28	0,07616%
Australia	2337	6,35676%
Austria	162	0,44065%
Bahrain	6	0,01632%
Belgium	249	0,67729%
Bermuda	561	1,52595%
Brazil	556	1,51235%
Canada	2134	5,80459%
Chile	187	0,50865%
China	640	1,74083%
Colombia	69	0,18768%
Cyprus	17	0,04624%
Czech Repu	29	0,07888%
Denmark	302	0,82146%
Egypt	73	0,19856%
Finland	317	0,86226%
France	1002	2,72549%

Germany	928	2,52421%
Greece	224	0,60929%
Hong Kong	470	1,27842%
Hungary	33	0,08976%
India	652	1,77347%
Indonesia	237	0,64465%
Ireland	327	0,88946%
Italy	468	1,27298%
Japan	4687	12,74888%
Jordan	9	0,02448%
Kazakhstan	9	0,02448%
Kuwait	49	0,13328%
Luxembourg	81	0,22032%
Malaysia	361	0,98194%
Malta	2	0,00544%
Mauritius	10	0,02720%
Mexico	226	0,61473%
Morocco	18	0,04896%
Netherland	474	1,28930%
New Zealand	222	0,60385%
Nigeria	8	0,02176%
Norway	198	0,53857%
Oman	25	0,06800%
Peru	38	0,10336%
Philippines	131	0,35633%
Poland	196	0,53313%
Portugal	116	0,31553%
Qatar	36	0,09792%
Saudi Arab	67	0,18224%
Singapore	379	1,03090%
South Africa	667	1,81427%
South Korea	716	1,94756%
Spain	504	1,37091%
Sri Lanka	8	0,02176%
Sweden	572	1,55587%
Switzerland	655	1,78163%
Taiwan	855	2,32564%
Thailand	221	0,60113%
Turkey	199	0,54129%
United Arab Emirates	35	0,09520%

United Kingdom	2919	7,93983%
United States	10055	27,35013%
Zimbabwe	8	0,02176%
Total	36764	100%
<i>Panel B. Distribution of the sample by geographic area</i>		
<i>Continent</i>	<i>Number observations geographic area</i>	<i>Percent (%)geographic area</i>
Africa	711	2,217%
America	13255	40,103%
Asia	9469	22.142%
Australia	2559	7,851%
Europe	10770	27,686%
Total	36764	100,00%

2.2 Dependent variable

To test our hypotheses, we used ESG performance data (ESG SCORE) from Asset4. This score is based on three dimensions: environmental, social and corporate governance. ESG performance measures a company's ability to reduce its environmental emissions, to use natural resources efficiently in production processes and to support research and development of eco-efficient products and services. Social performance measures a company's ability to generate the trust and loyalty of its employees, to respect the fundamental conventions of human rights, to be a good citizen, to protect public health, to respect ethics business and create value-added products and services.

Following Literature Dalal, & Thaker, (2019); Eliwa et al., (2019); Lopez de Silanes et al., (2019) our study measures the level of ESG performance by calculating the arithmetic mean of the three scores: social, environmental and corporate governance. The overall ESG SCORE is expressed as a percentage ranging from 0 to 100%. In order to capture the impact of board characteristics on ESG performance with the time it takes for the effects to appear and mitigate endogeneity issues, governance variables from any year are linked to the ESG measure of the following year.

2.3 Independent variables

The independent variables included in our econometric models are board size, board independence, and proportion of women on the board, sustainable compensation incentives and CSR sustainability committee.

Table 2. Variables measurement and the expected relationship with ESGSCORE

<i>Variable</i>	<i>Measurement</i>	<i>Expected sign with ESG SCORE</i>
ESG performance (ESGSCORE)	This score is based on three dimensions: environmental, social and corporate governance	Dependent variable
Board size (BOARDS)	Total number of directors on the bank's board	Positive
Board independence (BINDEP)	Percentage of independent board members as reported by the company (variable Independent Board Members)	Positive
Gender diversity (GENDERD)	Total number of women on the board of directors divided by the total number of board members	Positive

Sustainable Compensation incentives (SCINCENTIVES)	Dummy variable (yes = 1/no = 0) responding either the question “Does the Company have an extra-financial performance oriented compensation policy” or “Is the senior executive's compensation linked to CSR Sustainability target”.	Positive
CSR sustainability committee (CSRSUSTCOMM)	Dummy variable (yes = 1/no = 0) responding the question “Does the company have a CSR Sustainability committee or team?”	Positive

The next section will test the hypotheses regarding the effect of the board characteristics on the ESG SCORE, using panel data analysis. We estimate the following models to measure the relationship between ESG performance and independent variables:

$$\text{ESGSCORE}_{it} = \beta_0 + \beta_1 \text{BOARDS} + \beta_2 \text{BINDEP} + \beta_3 \text{GENDERD} + \beta_4 \text{SCINCENTIVES} + \beta_5 \text{CSRSUSTCOMM} + \varepsilon$$

2. EMPIRICAL RESULTS AND DISCUSSION

2.1 Descriptive Statistics

Panel A of Table 3 illustrates the descriptive statistics for the dependent and independent variables. The descriptive statistics table includes minimum, maximum, mean, and standard deviation. The average level of ESG performance (ESG SCORE) of the analyzed company is 50.86%, with a maximum equal to 98%. This reveals that corporate sustainability performance for the period 2002-2017 was highly satisfactory by the standards of the score definition. Board size is 10.38562 with a minimum of one member for every forty-four directors. Similarly, the average for board independence appears to be insufficient, given that some corporate boards do not have an independent director (a maximum of 100% with a minimum value of 0%). Similarly, the mass of women (the proportion of women on the boards) reaches an average value of (11%) and the maximum value is 100%, this still seems low but suggests that there are companies whose board members are all women. Similarly, the adoption average of Sustainable Remuneration incentives seems low considering that 16% of our observations have a Sustainable Remuneration incentive. In addition, around 46% of companies have adopted a sustainable remuneration policy.

The correlation matrix (Panel B Table 3) highlights very important relationships between the main variables of the study. Specifically, ESG SCORE is found to be positively associated with CSR sustainability committee, sustainability incentive compensation, board size, and gender diversity. These relationships show that more ethical and responsible companies tend to appoint more women to their boards of directors, more frequently a high level of independence of the board of directors, have adopted remuneration incentives for sustainable development and create a committee of sustainable development dedicated solely to sustainable development. Interestingly, a high proportion of board independence and CSR sustainability committee is positively associated with compensation for sustainability incentives, suggesting that companies whose boards are served by board independence high-level administration and created a CSR sustainability committee adopted a sustainability incentive remuneration strategy. On the contrary, the extent of board independence is negatively correlated with the size of the board.

Table 3. Summary statistics of ESG SCORE and explanatory variables.

<i>Panel A. Descriptive statistics</i>					
Variable	Obs	Mean	Std. Dev.	Min	Max
ESGSCORE	36764	50.86226	17.84807	6.042121	98.01675
BOARDS	36764	10.38562	3.72292	1	44
BINDEP	36764	54.54049	28.78914	0	100

GENDERD	36764	11.12622	14.64738	0	100
SCINCENTIVES	36764	.1615983	.3680869	0	1
CSRSUSTCOMM	36764	.4607496	.4984638	0	1

Panel B. Metrix of correlation

	ESGSCORE	BOARDS	BINDEP	GENDERD	SCINCENTIVES	CSRSUSTCOMM
ESGSCORE	1.0000					
BOARDS	0.2087	1.0000				
BINDEP	0.1507	-0.1268	1.0000			
GENDERD	0.3273	0.0035	0.1014	1.0000		
SCINCENTIVES	0.2491	0.0171	0.1837	0.0627	1.0000	
CSRSUSTCOMM	0.5484	0.1474	-0.0111	0.1746	0.2299	1.0000

This table shows the summary statistics of ESG SCORE and explanatory variables. The number of firm-year observations is 36.764 for all the variables. Data source: Thomson Reuters Asset4 and Datastream.

2.2 Regression results and discussion

Table 4 shows the results of estimating equation (1) that could satisfy our assumptions about the effect of board composition, activity, and compensation on ESG performance for a sample of 36,764 observations from 60 countries. Hypothesis 1 predicts that board size has a positive effect on company performance in ESG. In our model (Table 4), BOARDS are significant and positively correlated with ESG performance for the full sample set and for all different geographic regions. When it comes to interactions, the final sign of the predictor's effect is immediately readable from the grading table. In fact, coefficients associated with BOARDS have significant effects on plaque increase positively affecting ESG performance, and this finding is in line with studies (Harjoto, & Wang, 2020). Large boards are more likely to be served by managers with different skills and attitudes, including those with a strong penchant for a culture of sustainability. This result supports hypothesis 1.

Hypothesis 2 predicts a relationship between the number of independent board meetings and ESG performance, even if the score is certain. In other words, according to the existing literature, the number of meetings positively affects the performance of the ESG. Our results show a positive relationship: the coefficient associated with BINDEP is positive and significant across all geographical regions. This result is consistent with studies by Arayssi et al., 2020; Harjoto, & Wang, 2020). A large number of board meetings are likely to act as improvement elements that increase the company's performance in ESG. This result supports hypothesis 2.

Hypothesis 3 predicts a positive relationship between gender diversity and ESG performance. In this case, the hypothesis appears to be confirmed. In all models presented in (Table 4), gender is significant and positively correlated with ESG scores. Therefore, in line with the correlation results (see Table 4), the econometric model indicates that higher proportions of women on the board are associated with better sustainability performance. This positive effect is largely in line with the findings (Ajaz et al., 2020). A higher proportion of gender diversity is more likely to be served by women with different skills and attitudes, including managers distinguished with a strong penchant for a culture of sustainability. This result supports hypothesis 3.

In accordance with the prediction of our hypothesis 4, which predicts a positive relationship between executive compensation and the extent of sustainability performance, the results of the regression results show a positive and significant coefficient of the SCINCENTIVES variable.

Hypothesis 4 has been strongly verified. We therefore argue that in the presence of sustainable incentive compensation, the Board of Directors and the Compensation Committee may be in a better position to assess the company's carbon risks in order to design a more effective executive compensation system, which can improve the performance of the company's ESG. Therefore, any token compensation for

sustainability incentives, without the need for clear guidelines and mandatory emissions reduction targets, could improve the performance of a process-oriented ESG.

Finally, hypothesis 5 is strongly verified: the establishment of a CSR committee seems to positively affect the performance of companies in terms of ESG (see table 4). According to stakeholder theory, this observation confirms that the CSR committee helps banks to strengthen their credibility on sustainable development issues and to gain legitimacy with all stakeholders. This happens because the members of these committees have specialized CSR experience, skills and knowledge. Regarding this variable (see Table 4), in accordance with several studies, it is interesting to note that it has a positive and significant effect on ESG performance. These results show that the best performance in terms of sustainable development is obtained in particular by larger and more profitable companies (Alessandrini, & Jondeau, 2020).

Table 4. Results of regression models of ESG SCORE

ESGSCORE	Full set	Panel regression models of ESG SCORE by geographic area				
		Africa	America	Asia	Australia	EU countries
BOARDS	37.91*** (.7417379)	9.10*** (1.441503)	25.19*** (1.107434)	3.43*** (.08749)	21.82*** (2.702409)	25.20*** (.8212392)
BINDEP	24.35*** (.0666698)	5.10*** (.1435749)	21.57*** (.1364692)	8.55*** (.038479)	20.24*** (.2042467)	25.28*** (.1248584)
GENDERD	53.79*** (.2687777)	2.52** (.1548652)	16.59*** (.1618966)	93.19*** (.6098995)	9.35*** (.1590065)	2.06** (.0199053)
SCINCENTIVES	21.99*** (4.498195)	7.90*** (8.406084)	14.55*** (4.561186)	0.35 (.2148649)	7.06*** (3.741659)	12.10*** (4.375989)
CSRSUSTCOMM	109.34*** (16.5463)	7.25*** (11.88949)	66.04*** (17.05706)	49.30*** (11.55212)	21.81*** (11.77062)	51.48*** (14.65256)
Total	36764	711	13255	9469	2559	10770
F	4271.32	56.28	1503.46	3429.98	455.77	975.44
Prob > F	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
R-squared	0.4108	0.3242	0.4051	0.6850	0.5173	0.3522
Adj R-squared	0.4107	0.3184	0.4048	0.6848	0.5161	0.3519

This table shows the results regression for all set of a sample of 33764 observation and those by geographic areas. Data source: Thomson Reuters Asset4 and Datastream. Significant coefficients are indicated by the usual significance levels: ***, 1%; **, 5%; *, 10%

SUMMARY AND CONCLUSION

Building on previous literature demonstrating that boards of directors play a major oversight role in financial institutions, this study examines the relationship between board composition (size, activity, gender diversity, and compensation) and ESG performance in a large sample of 36,764 observations of 6000 listed companies for the period 2002 -2017. The main findings reveal that size, autonomy, gender diversity, compensation, and CSR positively influence company ESG performance for all full-sample group regression models for either geographic region. For this reason, we support the critical perspective that emphasizes board characteristics, such as board size, share of board independence, proportion of women on the board, executive sustainability incentive compensation, and a corporate social responsibility committee, which are very important to enhance company performance at ESG.

The empirical findings of this research provide a significant impact of environmental, social, and institutional controversies on ESG performance. This indicates that the importance of this variable to ESG performance will be more evident if the region is to implement stronger corporate governance measures. Since this variable is important in building a good reputation and encouraging the company to become more involved in activities and performance in social, environmental and governance, companies should strive to increase their ESG score. This study has important implications, especially for managers and regulators. From a management standpoint, our work suggests that directors and CEOs should pay more attention to board size, gender diversity, and the independence of their boards.

Also, to improve the performance of the ESG, it is important to appoint a CSR committee and to ensure a high level of compensation for sustainability incentives. Since larger boards positively impact ESG performance, company directors should select male and female directors to increase board size. Likewise, our study suggests evidence to support the impact of sustainability incentive compensation for all geographies as a strategic tool that demonstrates the company's strong commitment to ESG performance. Our study investigates the CRS, the ESG score, and the empirical evidence of our study sample showing that a CSR committee is more effective in improving CSR when it is characterized by a greater proportion of independent directors, than by a female. Furthermore, having a CSR Sustainability Committee, which works with the CSR Committee, can represent another strategic tool for maximizing the company's sustainability opportunities.

For regulators, our findings reveal the need to strengthen corporate governance principles for a sample of companies from multinational contexts in order to place special emphasis on board composition, executive sustainability incentives, and a CSR committee. However, our study presents some limitations. Our empirical analysis is based on the assumption that ESG performance is an effective measure of the sustainability performance of the study sample. It would be interesting to analyze the same relationship by adopting other measures of ESG performance, such as the Bloomberg ESG score. To date, however, data availability remains a problem for these investigations.

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